



SRI SHANMUGHA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution

Pullipalayam, Morur (Po.), Sankari (Tk.), Salem (Dt.) - 637 304.

QUESTION BANK

Subject Code & Name :	23AD410 & DATA EXPLORATION AND VISUALIZATION	
Faculty Name , Designation and Department	R.ASHA, AP / AI&DS	
Class : III	Semester : V	Academic Year: 2025-2026

UNIT I: EXPLORATORY DATA ANALYSIS

Q. No	5 x 2 = 10 Marks	COx	Kx
1.	List down the steps in EDA.	CO1	K1
2.	Recall the different categories of data available in EDA.	CO1	K1
3.	Define Bayesian Analysis.	CO1	K1
4.	Outline the software tools available for EDA.	CO1	K2
5.	Mention any 4 transformation techniques.	CO1	K1
Q.No	4 x 16 = 64 Marks	COx	Kx
1.	Compare Exploratory Data Analysis (EDA) with Classical and Bayesian analysis approaches in terms of methodology, assumptions and applications with suitable examples..	CO1	K2
2.	(a) Explain the software tools available for EDA. (8 marks) (b) Explain in detail the visual aids for EDA. (8 marks)	CO1	K2
3.	Explain the various transformation techniques in EDA.	CO1	K2
4.	Explain in detail about the grouping and aggregation techniques in EDA using pivot tables and cross tabulations with suitable example for each.	CO1	K2



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UNIT II: VISUALIZING USING MATPLOTLIB

Q. No	5 x 2 = 10 Marks	COx	Kx
1.	List down the steps involved in line plot.	CO2	K1
2.	Compare and contrast Bivariate and Univariate data using seaborn and pandas.	CO2	K2
3.	Define the term Matplotlib.	CO2	K1
4.	Show histogram is always a bar chart.	CO2	K1
5.	State any 4 map projections.	CO2	K1
Q.No	4 x 16 = 64 Marks	COx	Kx
1.	Illustrate the process of creating three-dimensional plots using Matplotlib, highlighting its significance in visualizing complex data structures.	CO2	K2
2.	Describe the steps involved in visualizing geographic data using Basemap, emphasizing its applications in spatial data analysis.	CO2	K2
3.	Summarize different types of visualizations available in Matplotlib such as line plots, scatter plots, histograms, and contour plots with appropriate examples.	CO2	K2
4.	(a) Discuss how Seaborn helps to visualize the statistical relationships. (8 marks) (b) Discuss Seaborn with code and examples. (8 marks)	CO2	K3

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UNIT III: UNIVARIATE ANALYSIS

Q. No	5 x 2 = 10 Marks	CO_x	K_x
1.	Mention the reason to reduce in-equality.	CO3	K1
2.	Define Lorenz Curve.	CO3	K1
3.	Compare scaling and standardizing.	CO3	K2
4.	Interpret the meaning of the term “scaling” in the context of uni-variate analysis.	CO3	K2
5.	State the statistical meaning of “level” and “spread.”	CO3	K1
Q.No	4 x 16 = 64 Marks	CO_x	K_x
1.	(a) Mention any 5 Essential Numerical Summaries in Statistics with example. (8 marks) (b) State the reason why you should Normalize / Standardize / Re scale your data. (8 marks)	CO3	K1
2.	Assume you are analyzing monthly sales data for a retail company that shows fluctuations due to seasonal effects. To better identify the underlying trend, you decide to smooth the time series data. Describe the techniques you would use for smoothing and explain the process with suitable examples.	CO3	K3
3.	You have been given a data set containing information about the daily temperatures recorded in a city over a year. Before performing any complex analysis, you are asked to conduct a uni-variate analysis on this single variable. Explain how you would approach this task and discuss the types of distributions and variables you might encounter.	CO3	K3

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4.	Provide a comprehensive guide to analyzing contingency tables with a focus on marketing data, including scenario-based questions.	CO3	K3
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UNIT IV: BIVARIATE ANALYSIS

Q. No	5 x 2 = 10 Marks	COx	Kx
1.	Recall the effect of transformations on the relationships between two variables.	CO4	K1
2.	Define resistant lines.	CO4	K1
3.	List three common methods used in bivariate analysis	CO4	K1
4.	Outline the difference between univariate and bivariate data.	CO4	K2
5.	Infer the role and importance of scatter plots in bivariate analysis.	CO4	K2
Q.No	4 x 16 = 64 Marks	COx	Kx
1.	Discuss in detail about the construction and interpretation of percentage tables and contingency tables in bivariate analysis.	CO4	K3
2.	Describe the use of scatterplots in visualizing relationships between two variables. Also explain how resistant lines are added and interpreted with suitable examples.	CO4	K2



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3.	(a) Discuss about contingency tables and their statistical significance. Illustrate their use through a market segmentation analysis example. (8 marks) (b) Discuss how the insights assist in targeting marketing efforts more effectively. (8 marks)	CO4	K3
4.	Explain the concept of transformations in bivariate analysis and how they can help in understanding complex relationships between variables.	CO4	K2

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UNIT V: MULTIVARIATE AND TIME SERIES ANALYSIS

Q. No	5 x 2 = 10 Marks	COx	Kx
1.	Interpret the concept of grouping within time series data	CO5	K2
2.	State the meaning and structure of a three-variable contingency table.	CO5	K1
3.	List the characteristics of time series data.	CO5	K1
4.	Outline the concept and purpose of longitudinal data analysis	CO5	K2
5.	Define Resampling	CO5	K1
Q.No	4 x 16 = 64 Marks	COx	Kx
1.	Discuss the fundamentals of Time Series Analysis (TSA) and its applications in understanding temporal data patterns, providing examples to illustrate its significance.	CO5	K3
2.	Describe the process of visualizing multivariate data and grouping in the context of time series analysis, highlighting their importance in identifying patterns and trends.	CO5	K2
3.	Apply the concept of multivariate analysis to explain the role of a third variable in data interpretation. How can causal explanations be used to better understand multivariate relationships?	CO5	K3
4.	(a) Explain how to clean time series data, perform time-based indexing and apply resampling techniques. (8 marks) (b) Explain how each of these steps contributes to preparing the data for effective analysis. (8 marks)	CO5	K2